

National University of Science & Technology

Institute of Geographical Information Systems

Course Title: Digital Mapping and Image Processing

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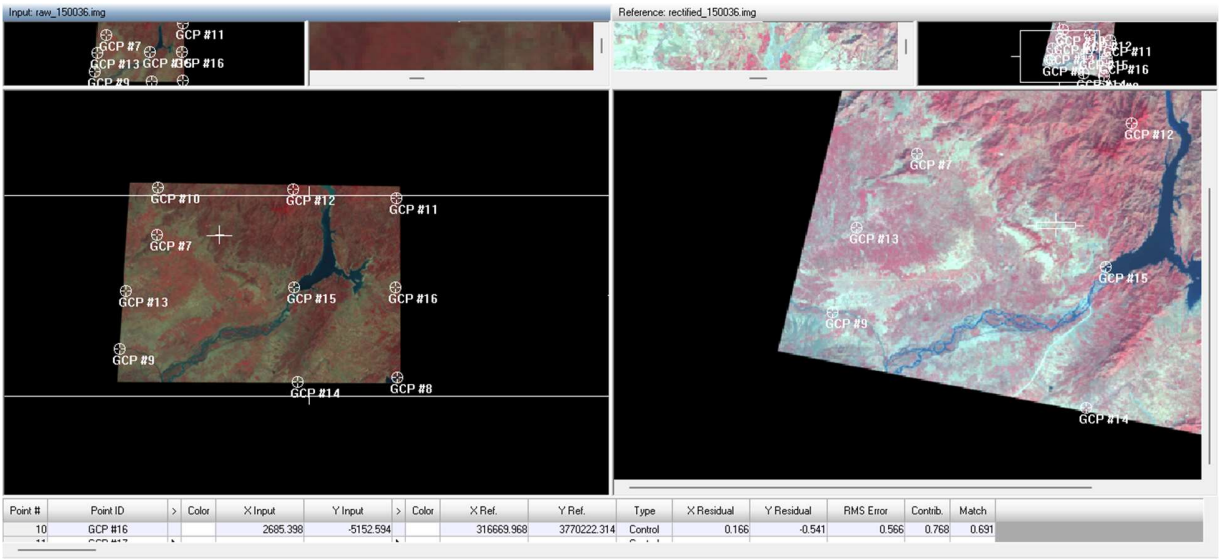


Figure 1 GCP on image

Point #	Point ID	Color	X Input	Y Input	Color	X Ref.	Y Ref.	Type	X Residual	Y Residual	RMS Error	Contrib.	Match
1	GCP #7		379.609	-4647.857		250355.147	3797738.053	Control	-0.124	0.602	0.615	0.835	0.534
2	GCP #8		2705.470	-6020.650		312479.668	3744239.844	Control	-0.309	0.561	0.641	0.870	0.211
3	GCP #9		24.580	-5747.229		233699.061	3766807.553	Control	1.360	-0.060	1.361	1.848	0.209
4	GCP #10		391.394	-4190.379		253216.971	3811314.789	Control	-0.435	0.055	0.438	0.595	0.634
5	GCP #11		2698.686	-4291.370		321821.683	3795878.433	Control	0.281	0.208	0.349	0.474	0.789
6	GCP #12		1698.674	-4208.148		292307.436	3803741.051	Control	0.085	-0.014	0.086	0.117	0.556
7	GCP #13		83.710	-5185.316		238498.502	3783265.445	Control	-0.891	-0.326	0.949	1.288	0.063
8	GCP #14		1744.494	-6063.436		283432.517	3748124.950	Control	-0.734	0.090	0.740	1.005	0.392
9	GCP #15		1702.605	-5151.099		287241.275	3775555.016	Control	0.601	-0.575	0.832	1.130	0.221
10	GCP #16		2685.398	-5152.594		316669.968	3770222.314	Control	0.166	-0.541	0.566	0.768	0.691
11	GCP #17							Control					

Figure 2 RMS Errors

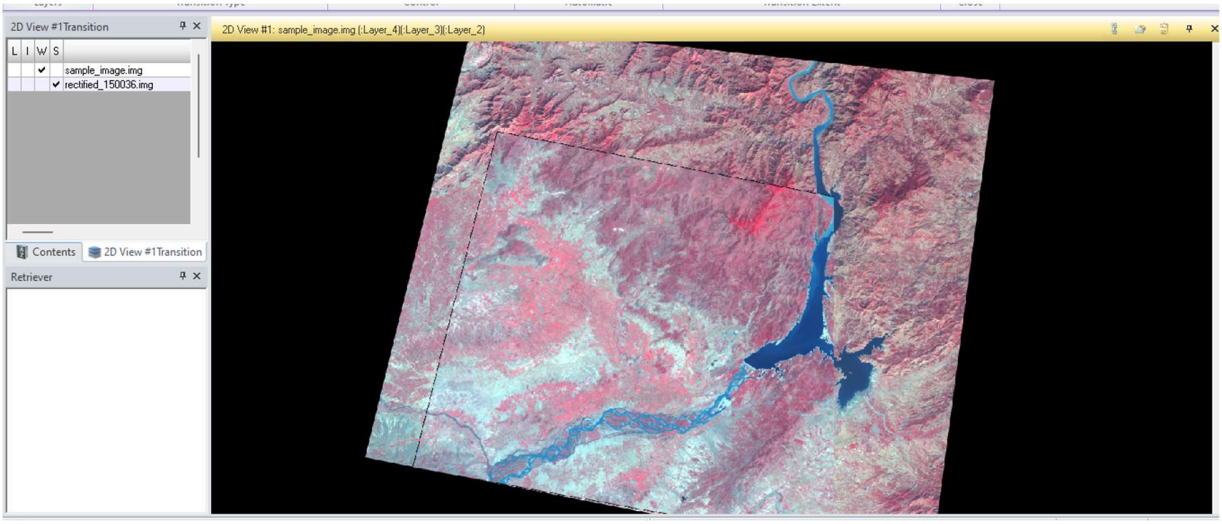


Figure 3 Final Result

1. What is GCP and what criteria should be for the selection of a good GCP?

A **GCP (Ground Control Point)** is a clearly identifiable point on the ground whose exact location (coordinates) is known. It's used in mapping, surveying, and remote sensing to align images or maps with real-world positions.

Criteria for a good GCP:

- **Clearly visible** in imagery (e.g., road intersections, corners of buildings).
- **Stable and permanent** (not something that can move or change).
- **Well-distributed** across the study area (not clustered in one spot).
- **Accurately measured** using GPS or total station.
- **Sharp and unambiguous** (easy to identify the exact point).

2. What is the minimum number of GCP necessary for 1st order polynomial transformation?

3 GCPs are required (since affine transformation has 6 unknowns, and each GCP gives 2 equations).

3. What is RMS Error?

- It is the measure of the average difference between the actual (measured) GCP coordinates and the transformed coordinates.
- Lower RMS = higher accuracy.

4. What are RMS errors from Control Points _____ and Check Points_____?

- **From Control Points:** Show how well the transformation fits the GCPs used in the calculation.
- **From Check Points:** Show the independent accuracy of the transformation using extra points not included in the calculation (better indicator of true accuracy).

5. Why are we using Nearest Neighbour resampling? What advantage does it have over other methods, bilinear and cubic convolution?

<u>Nearest Neighbour Resampling:</u>

- It assigns the value of the closest pixel to the new pixel position, without averaging or interpolation.

Advantages over Bilinear and Cubic Convolution:

- **Preserves original pixel values** (no alteration of radiometric data).
- **Best for categorical/discrete data** (like land cover classes, thematic maps) where values must stay unchanged.
- **Fastest method** (computationally less expensive).

In contrast, **bilinear** and **cubic convolution** smooth values by averaging, which is good for continuous data (like imagery) but distorts categorical data.

6. What kind of differences you can see in both images?

Water Body (Shrinking):

- The reservoir/river in the newer image looks narrower in some stretches compared to the older one → this suggests a **reduction in water extent**.

Vegetation & Land Cover:

- The reddish areas (vegetation in false color composite) show variation — some patches look denser, others reduced.
  This suggests **change in vegetation cover** over time.